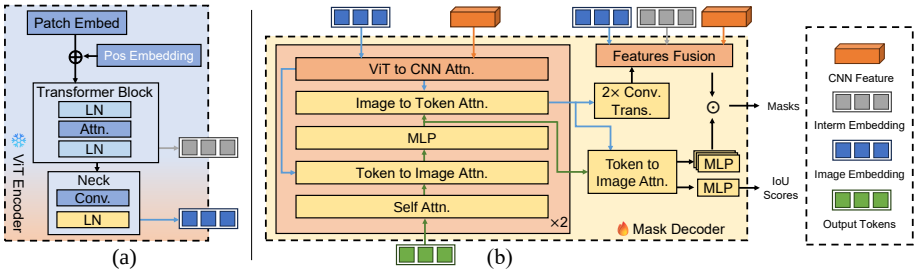

Fig. 2: Detailed architecture of ViT Encoder and Mask Decoder. (a) represents the ViT encoder, while (b) showcases the lightweight decoder of SAM. The CNN feature is derived from the output of the CNN encoder. The yellow and blue modules represent the original SAM structure.

that the image encoder of SAM is not able to capture enough features effectively from unseen endoscopy images. To address this issue, the CFA module is designed to learn multi-scale features and multi-level representations, thereby enhancing the feature extraction capabilities of the encoder.

Firstly, to achieve automatic segmentation, we modified the architecture of SAM by removing its prompt input and prompt encoder components while preserving its image encoder and mask decoder parts. Recent studies[17] have demonstrated that ViT is more focused on low-frequency signals, while CNN is more adept at processing high-frequency signals. Hence, we integrate a parallel CNN-based branch to compensate for the absence of high-frequency and local features. Furthermore, we augment the mask decoder of SAM by proposing an additional multi-head cross-branch attention block to facilitate the integration of features extracted from both the ViT encoder and the CNN encoder. For the features $\mathbf{F}_v$ from the ViT branch and $\mathbf{F}_c$ from the CNN, the cross-branch attention can be formulated as follows:

$$\text{CrossBranchAttention}\left(\mathbf{F}_v, \mathbf{F}_c\right) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathsf{T}}}{\sqrt{d}}\right)\mathbf{V}. \tag{1}$$

where $\mathbf{Q} = \mathbf{F}_v\mathbf{W}^{\mathbf{Q}}$, $\mathbf{K} = \mathbf{V} = \mathbf{F}_c\mathbf{W}^{\mathbf{K}}$, and d is the number of channels of each head of $\mathbf{F}_v$. Considering that CNN features offer more precise position information, we substituted SAM's original position embedding in the mask decoder with the final output features from the CNN encoder. Furthermore, we integrated the cross-branch attention mechanism into the attention block of the mask decoder, repeating this process twice to ensure integration of multi-scale features from both the ViT and CNN encoders, as shown in Fig. 2.

Secondly, to obtain more precise segmentation results, we integrate high-level context and low-level boundary information from the encoder with the decoder features of SAM to augment the output information. Specifically, we combine the shallow local features obtained from the intermediate embedding of the ViT encoder, the final global features obtained from the image embedding

of the ViT encoder, and the final features of the CNN encoder, as illustrated in Fig. 2. This approach fully harnesses the rich edge information, extensive global context information, and local position details of each encoder branch. Thus, we can effectively integrate multi-level features from both ViT and CNN.

2.2 Uncertainty-guided Prediction Regularization Module

To augment the generalization capability of SAM, we propose a novel training strategy involving the selective activation of LayerNorm within the encoder. We also employ the ground truth as a 'hint' to further guide the training process by correcting the confidence.

Given that SAM is trained on natural data, its performance may deteriorate in polyp images due to the domain transfer. As previously suggested [13], it is a particularly effective technique for domain transfer by adjusting the normalization layer. Despite the introduction of LayerNorm [1] potentially reducing training time, it fundamentally alters the distribution of the input data. When transferring SAM from natural images to endoscopy images, there is a shift in both the data distribution and the corresponding feature space distribution. These distributional differences can induce internal covariate shifts, thereby influencing the model's performance. To improve the SAM's generalization in the endoscopy field, we fine-tune the normalization layer of the encoder. In this process, the model effectively adapts the data distribution in the target domain and mitigates the effects of internal covariate shifts.

Specifically, the LayerNorm of SAM's ViT encoder is divided into (1) Transformer block norm, and (2) neck layer norm, as illustrated in Fig. 2(a). Given that the features in the neck layer are closer to the output features of the encoder, we ultimately decided to train the neck layer normalization, which is equivalent to re-normalizing the features of the pre-trained ViT encoder. Coincidentally, in this work[32], the straightforward technique of adjusting normalization layers can surprisingly yield comparable or even superior performance to the robust baseline of fine-tuning all parameters.

Moreover, previous research[16] has demonstrated that predictions with lower uncertainty tend to exhibit superior out-of-distribution (OOD) performance, which is also beneficial for domain adaptation. SAM generates an IoU score output, which inherently can represent uncertainty (or, conversely, confidence). However, during the prediction process, SAM may frequently produce incorrect predictions for unseen data with high confidence, which is undesirable. To mitigate this problem, we strive to reduce the uncertainty of the model during training (i.e., to increase confidence). Inspired by [4], we utilize the ground truth as a hint to guide the learning of the model. First, we represent the IoU score of SAM as the image-level confidence c_i. Then we calculate the pixel-level confidence c_p to refine the uncertainty of each pixel using Eq. 2, where $\mathbf{U}_p \in \mathbb{R}^{B\times 1\times H\times W}$.

$$c_p = \left(1 - \frac{1}{H \times W} \sum_{i}^{H} \sum_{j}^{W} \mathbf{U}_p\right). \tag{2}$$

The term $\mathbf{U}_p$ represents the pixel uncertainty, defined as $\mathbf{U}_p = 1 - \sigma(|\mathbf{P}|)$. Here, σ represents the Sigmoid function, while $\mathbf{P}$ represents the output prediction. The final confidence is calculated as the sum of the image-level confidence and the pixel-level confidence, expressed as $c = \frac{1}{2}(c_i + c_p)$. This confidence is determined by the Bernoulli distribution, which decides whether to utilize the ground truth as a hint. In other words, if the confidence is low enough, we believe that the model requires a specific answer hint to learn the correct mask prediction. Thus, the answer is required as a hint, otherwise, it is not necessary. The weight of the hint is determined by the confidence c, which is expressed as follows:

$$\mathbf{P}' = c \cdot \mathbf{P} + (1 - c) \cdot \mathbf{Y}. \tag{3}$$

However, by minimizing the loss function, the model will tend to make $c = 0$ so that $\mathbf{P}'$ will always be GT. This means that the model does not actually learn. Therefore, a confidence loss is introduced to supervise c, which will increase when $c \to 0$, and the confidence loss is defined as follows:

$$\mathcal{L}_c = -log(c). \tag{4}$$

The final loss function is the sum of the segmentation loss $\mathcal{L}_s$ and the confidence loss $\mathcal{L}_c$, as defined in Eq. 5. Here, λ represents a hyperparameter. Specifically, the segmentation loss employed is a combination of CE loss, Dice loss, and MSE loss as $\mathcal{L}_s = \mathcal{L}_{ce} + 0.5 \cdot \mathcal{L}_{dice} + \mathcal{L}_{mse}$.

$$\mathcal{L} = \mathcal{L}_s + \lambda \mathcal{L}_c. \tag{5}$$

3 Experiments

Datasets. We conduct extensive experiments on five polyp segmentation datasets following PraNet[6], including Kvasir-SEG[10], CVC-ClinicDB[2], CVC-ColonDB[20], ETIS[19] and EndoScene[21]. Specifically, the training set consists of 900 images from Kvasir-SEG and 550 images from ClinicDB. The test sets comprise 100 images from Kvasir-SEG, 62 images from CVC-ClincDB, 380 images from CVC-ColonDB, 60 images from EndoScene, and 196 images from ETIS.

Implementations. We use PyTorch to implement our method and conduct experiments on a single NVIDIA RTX3090 GPU. The AdamW optimizer is utilized for training 16K iterations with a learning rate of 1e-5, a weight decay of 1e-4, and a batch size of 4. The CNN model we utilize is MSCAN-L, sourced from SegNeXt[7]. The input image size for the ViT branch is 1024×1024, while the input size for the CNN branch is 320×320. In the evaluation stage, we use two common metrics in medical image segmentation, *Dice* and *IoU*.

Results and Analysis. We compare our method with several state-of-the-art polyp segmentation methods and some SAM-based methods in Table 1. It is evident that while the performance enhancement on Kvasir and CVC-ColonDB is